



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CANDY MATCH SAGA: CANDY AI LAB

An Intelligent Rule-Based Candy Game with Player Analytics

“Where Every Match Teaches AI Something New.”

CSA0920 – Programming in Java

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QUICK RECAP

Project Concept:

- Candy Match Saga is an intelligent rule-based candy matching game.
- Players swap candies to create valid matches and earn points.
- The game supports Match 2, Match 3, Match 4 and Match 5 combinations.
- Custom candy synthesis and game rules can be created dynamically.
- Player performance is monitored using score, moves, matches and analytics.

Five Planned Modules:

1. Game Setup & Level Management
2. Candy Board & Match Processing
3. Candy Synthesis & Rule Management
4. AI Opponent & Adaptive Strategy
5. Player Analytics & Data Management

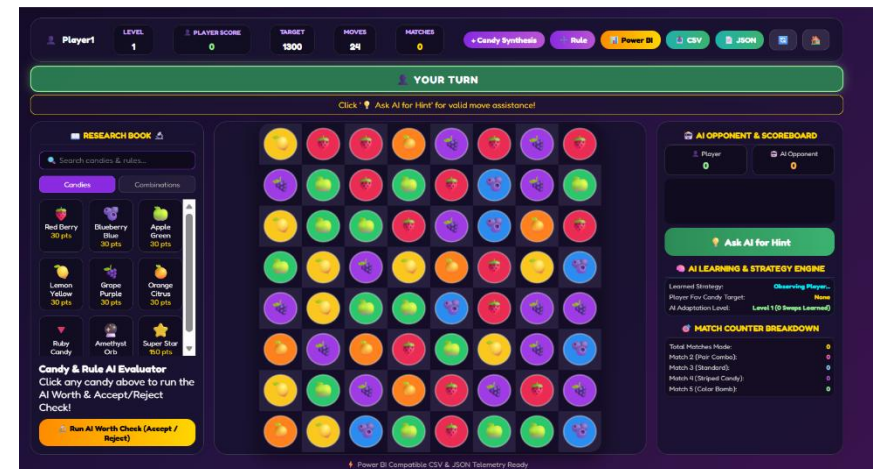
IMPLEMENTATION STATUS SUMMARY



Modules	Status	Approx. Completion
Game Setup & Level Management	Completed	100%
Candy Board & Match Processing	Completed	100%
Candy Synthesis & Rule Management	Completed	100%
AI Opponent & Adaptive Strategy	Completed	100%
Player Analytics & Data Management	Completed	100%

Overall Status: 100% Completed

- 20 levels implemented
- Rule-Based AI integrated
- Custom candies and rules implemented
- Analytics data generated
- CSV/JSON & Power BI integration completed



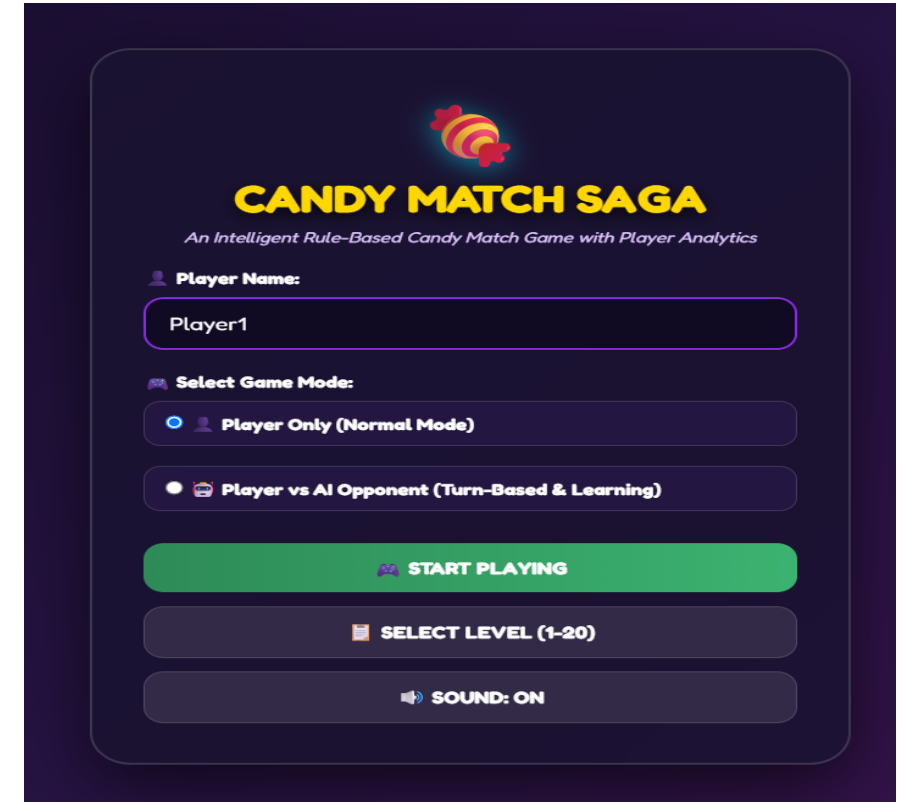
MODULE 1: Game Setup & Level Management

Implemented Features:

- Player name can be entered before starting the game.
- Supports Player Only and Player vs AI modes.
- Players can select levels from 1–20.
- Sound can be enabled or disabled.
- Initializes the game according to the selected player and level.

Key UI:

Player Name → Game Mode → Start Playing →
Select Level → Sound



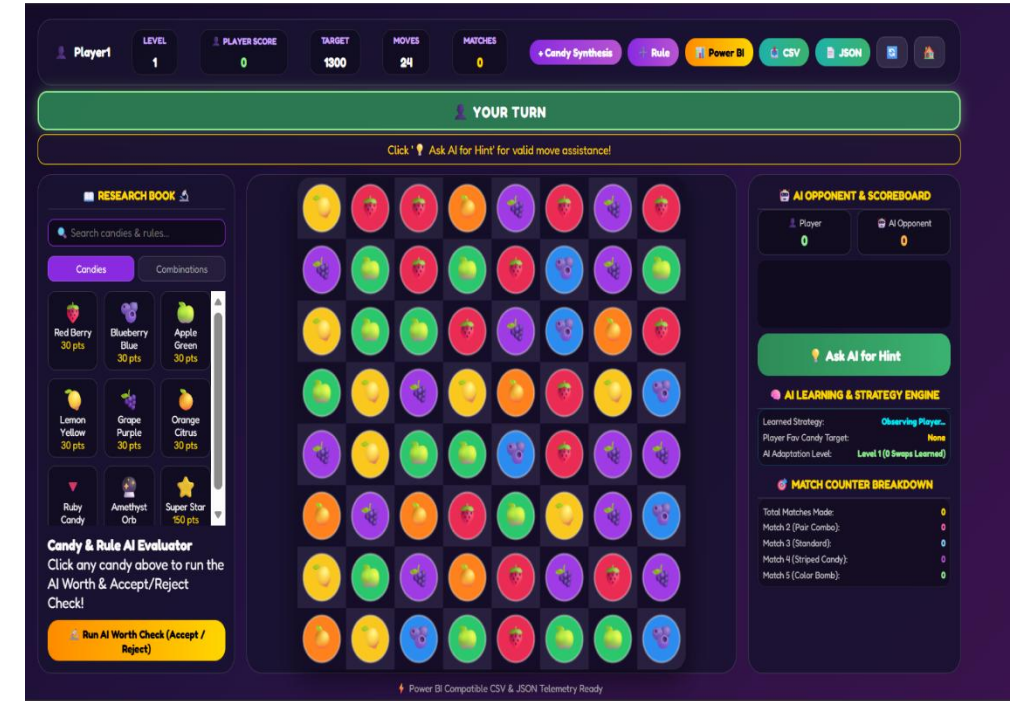
MODULE 2: Candy Board & Match Processing

Implemented Features:

- Displays a structured candy grid for gameplay.
- Supports multiple candy types such as Red Berry, Blueberry, Apple, Lemon, Grape and Orange.
- Detects valid candy combinations.
- Supports Match 2, Match 3, Match 4 and Match 5.
- Updates score, moves, target and match counters during gameplay.

Example:

Match 2 → Pair Combo
Match 3 → Standard Match
Match 4 → Striped Candy
Match 5 → Color Bomb



MODULE 3: Candy Synthesis & Rule Management



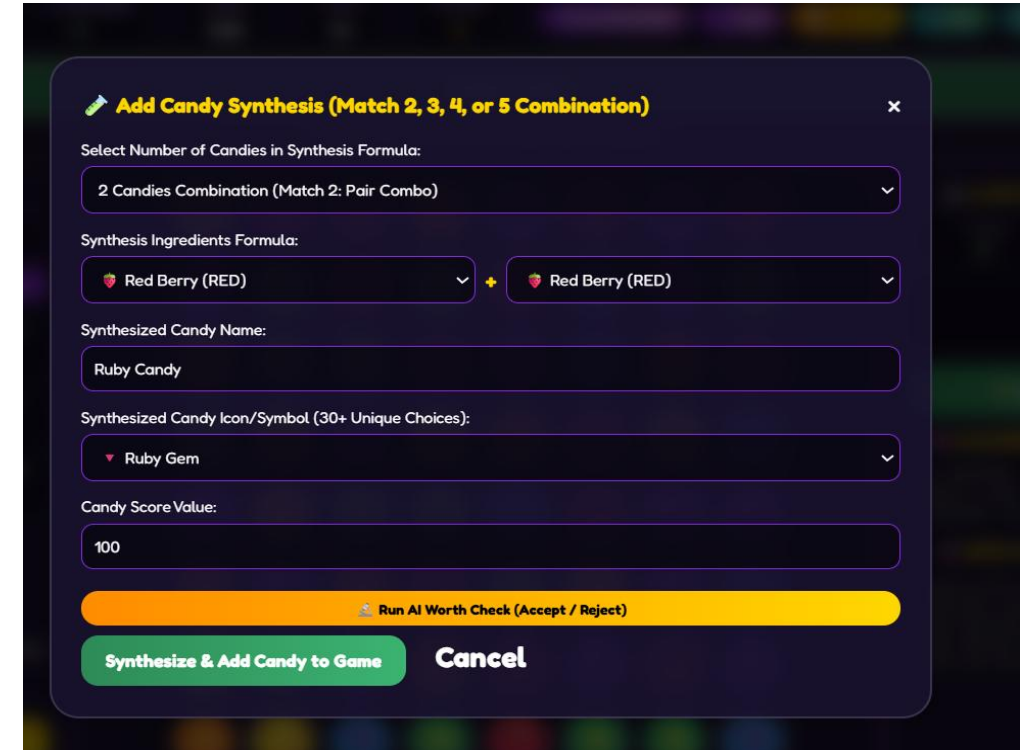
Implemented Features:

- Allows synthesis using 2, 3, 4 or 5 candies.
- User can select ingredients for a synthesis formula.
- Allows creation of custom synthesized candy names and symbols.
- Supports custom game rules such as Ruby Combination Bonus.
- AI Worth Check evaluates whether a candy/rule should be accepted or rejected.

Example:

Red Berry + Red Berry → Ruby Candy (+100 pts)

Lemon + Red Berry + Ruby Candy → Super Star (+150 pts)



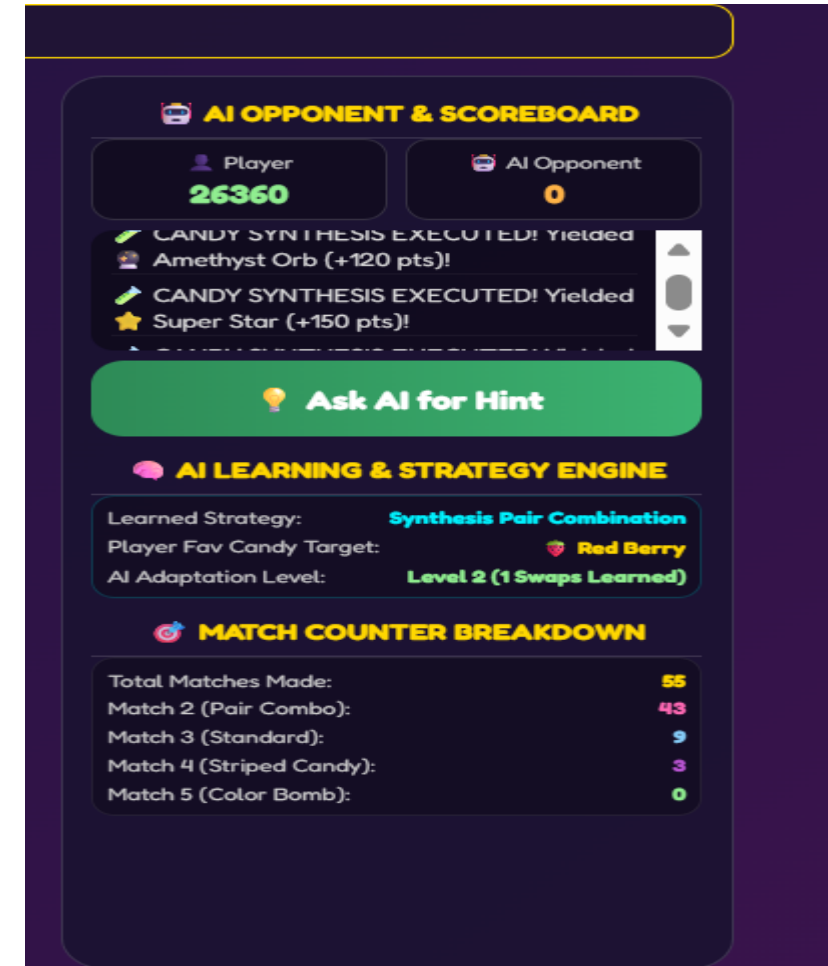
MODULE 4: AI Opponent & Adaptive Strategy

Implemented Features:

- Provides **AI-based hints** for valid player moves.
- Supports **Player vs AI Opponent** mode.
- Tracks player candy preferences and gameplay behaviour.
- Learns successful player strategies.
- Dynamically adapts the AI level based on observed player actions.

AI Learning & Strategy Engine:

- Learned Strategy: Synthesis Pair Combination
- Player Favourite Target: Red Berry
- AI Adaptation Level: Level 2 – Swaps Learned



MODULE 5: Player Analytics & Data Management

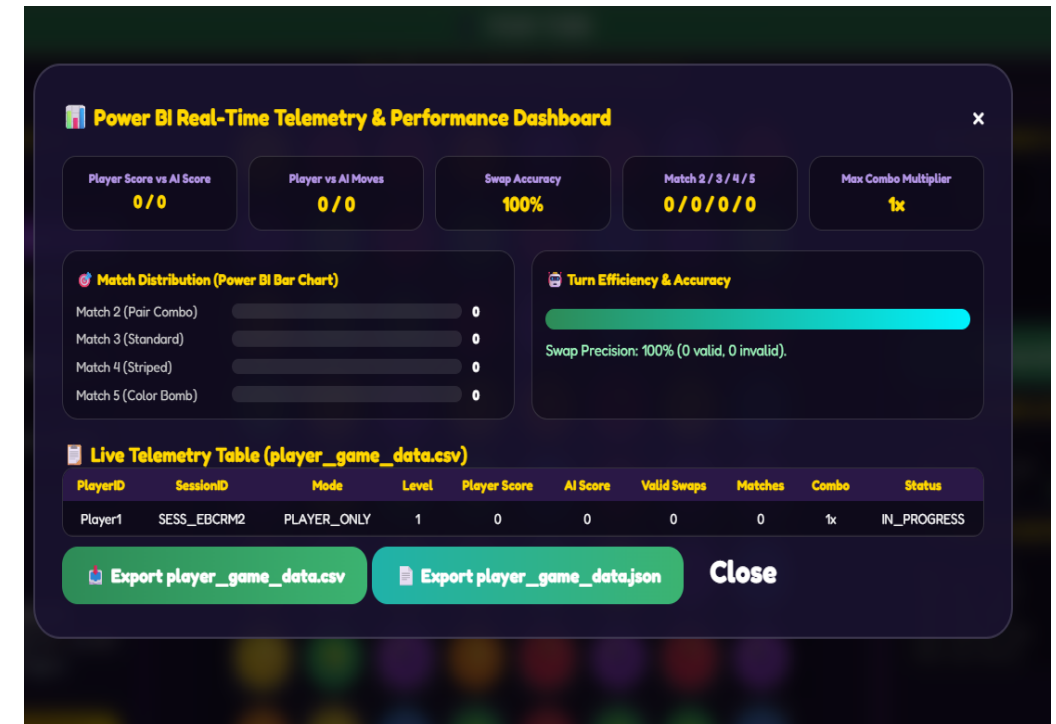


Implemented Features:

- Tracks player score and AI score.
- Records valid swaps and total matches.
- Categorizes matches into Match 2, 3, 4 and 5.
- Provides real-time performance visualization through the dashboard.
- Supports **CSV and JSON data export**.

Analytics Displayed:

- Player Score vs AI Score
- Player vs AI Moves
- Swap Accuracy
- Match Distribution
- Combo Multiplier
- Match Counter Breakdown



COLLECTION FRAMEWORK – FINAL USAGE

Collection	Usage in Game
ArrayList / List	Store match history and game events
HashMap / Map	Store candy names, symbols and score values
Array / 2D Array	Represent the candy game board
List	Store valid moves / matched candies
Map	Store player statistics and analytics data

Why Collections?

Collections provide efficient storage and retrieval of dynamic game data such as candies, matches, player statistics and rules.

MULTITHREADING FINAL IMPLEMENTATION

- Background game tasks can be executed independently without blocking the main game flow.
- Timer and periodic game updates can run separately from player interactions.
- AI decision-making can be processed independently when required.
- Background telemetry/data updates can be handled without interrupting gameplay.
- This improves game responsiveness and provides smoother user interaction.

EXCEPTION HANDLING FINAL IMPLEMENTATION



Exception Handling:

- Input validation is performed for player and game configuration.
- Invalid candy/rule combinations can be handled safely.
- Invalid numerical values are prevented from causing application failure.
- Data export errors can be handled using exception handling.
- try-catch blocks prevent unexpected errors from crashing the application.



JDBC / DATABASE INTEGRATION FINAL



Status: Completed

- Database structure finalized for player and game-session data.
- JDBC integrated to connect the Java application with MySQL.
- Player details, level, score and gameplay statistics are stored in the database.
- Match history and player performance data can be retrieved for analytics.
- Database integration provides persistent storage for player records and gameplay data.

LIVE DEMO

1. Start Game
- ↓
2. Select Game Mode & Level
- ↓
3. Perform Candy Swaps
- ↓
4. Generate Matches
- ↓
5. Synthesize Special Candy
- ↓
6. Use AI Hint / AI Opponent
- ↓
7. View Player Analytics
- ↓
8. Export CSV / JSON Data

TESTING SUMMARY



Test Case	Expected	Actual	Status
Board Initialization	Board created	Board created	Pass
Match-3 Detection	Candies removed	Candies removed	Pass
Special Candy	Created correctly	Created correctly	Pass
Invalid Move	Move rejected	Move rejected	Pass
AI Hint	Valid move shown	Valid move shown	Pass
Custom Rule	Invalid rule rejected	Rejected	Pass
Data Export	CSV/JSON generated	Generated	Pass
Analytics	Data displayed	Data displayed	Pass

Testing Result: Major functional test cases passed successfully.

ISSUES FACED ACROSS THE PROJECT & HOW RESOLVED

Issue 1 – Match Detection

Problem: Multiple match types required different validation conditions.

Solution: Implemented separate rule conditions for Match 2, Match 3, Match 4 and Match 5.

Issue 2 – Dynamic Candy Synthesis

Problem: Different ingredient combinations produce different special candies.

Solution: Created a configurable synthesis and rule system instead of hard-coding every combination.

Issue 3 – AI Adaptation

Problem: AI needed to respond to changing player behaviour.

Solution: Player moves, favourite candy targets and successful strategies are tracked for adaptive decision-making.



DEVIATIONS FROM REVIEW PLAN



Changes Made:

- Added Candy Synthesis as an interactive feature.
- Added Custom Game Rule creation.
- Added AI Worth Check for candies and rules.
- Added AI Learning & Strategy Engine.
- Added Power BI-style real-time telemetry dashboard.
- Added CSV and JSON export for gameplay analytics.

Reason:

The scope was extended to make the game more intelligent, configurable and analytics-oriented.

LEARNING OUTCOMES/CONCLUSION



Learning Outcomes:

- Improved Java OOP and modular programming skills.
- Learned to use Collections Framework for game data.
- Implemented Rule-Based AI and game logic.
- Gained experience in Exception & File Handling.
- Learned Player Analytics and Power BI integration.

Conclusion:

Candy Match Saga successfully combines Java game development, Rule-Based AI, custom game rules and player analytics into a modular application.

Future Improvement:

- Improve AI move evaluation.
- Add more advanced gameplay features.
- Enhance analytics and visualization.
- Improve overall game performance and user experience.



Engineer to Excel



THANK YOU